



Paper 21 summary

Desde Microsoft CMT <noreply@msr-cmt.org>

Fecha Sáb 2026-05-16 19:34

Para Claudia Fernanda Rodriguez Rodriguez <cfrodriguez@inm.gov.co>

No suele recibir correo electrónico de noreply@msr-cmt.org. [Por qué es esto importante](#)

Hello.

Here is submission summary.

Track Name: LoHacemosxTIC2026

Paper ID: 21

Paper Title: Development of a Prototype Measurement System for Comparing Atomic Oscillator Signals Using GNSS.

Abstract:

Time and frequency transfer using GNSS systems constitutes a fundamental tool for ensuring synchronization and temporal traceability in scientific, industrial, and telecommunications applications. In this context, National Metrology Institutes require measurement systems capable of performing comparisons between atomic oscillators, enabling the generation and maintenance of national UTC(k) time scales, as well as the provision of high-accuracy measurement services. This work describes the development of a prototype measurement system for the comparison of atomic oscillator signals using the satellite-based Common-View technique with a low-cost GNSS receiver. The project was developed through the collaboration between the groups GICOG and GIMCI, integrating expertise in technological development, data processing, and time and frequency metrology. The implemented prototype integrates a low-cost GNSS receiver, an external reference provided by an atomic oscillator, and software for GNSS data acquisition and processing. The system enables the capture of raw observations and their conversion into standard formats. Algorithms were developed for observation processing and time correction procedures, allowing the generation of CGGTTS files.

System validation was carried out by comparing the obtained results with those from the reference system developed by NIST, both connected to the same reference. Statistical tools such as error calculation, Type A uncertainty, normalized error E_n , and correlation analysis were used for the evaluation. The results demonstrated that the developed system satisfies the criterion of maintaining the time scale within ± 100 ns with respect to UTC, obtaining an error of (-3.01 ± 0.55) ns. Furthermore, the compatibility analysis yielded a value of $E_n = -0.77$, indicating compatibility between both systems, while the correlation coefficient of 0.98 showed a strong relationship between the prototype results and the reference system.

Created on: Sun, 17 May 2026 00:12:53 GMT

Last Modified: Sun, 17 May 2026 00:30:01 GMT

Authors:

- cfrodriguez@inm.gov.co (Primary)
- nbahamon@inm.gov.co
- neverap@udistrital.edu.co

Secondary Subject Areas: Not Entered

Submission Files:

Abstract Prototype Measurement Sysytem.pdf (232 Kb, Sun, 17 May 2026 00:10:10 GMT)

Submission Questions Response:

1. Topics

Technology and artificial intelligence

2. Additional Information

Author 1: Specialist in project management, Student/Researcher, +57 3125532417

Author 2: MSc in applied mathematics, Researcher, +57 3196766829

Author 3: PhD in Engineering, Professor/Researcher, +57 3012401090

3. I am interested in having the work submitted for publication in a scientific journal

Agreement accepted

4. Registration Benefit

Master's Students

Thanks,
CMT Team.

Please do not reply to this email as it was generated from an email account that is not monitored.

To stop receiving conference emails, you can check the 'Do not send me conference email' box from your User Profile.

Microsoft respects your privacy. To learn more, please read our [Privacy Statement](#).

Microsoft Corporation
One Microsoft Way
Redmond, WA 98052